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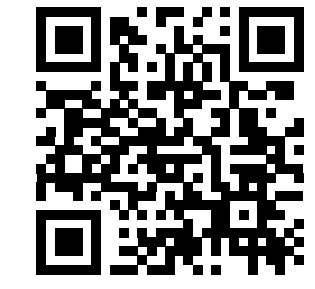
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# Universal Test-Time Training

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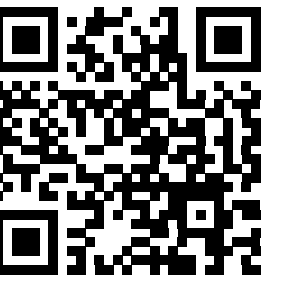
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Paper



Website



Code

## Motivation and approach

### 1 Memory need not be partitioned by layer

TTT stores context in online fast weights. Layer-private memory limits each layer to its own budget and writes. uTTT shares a dense state or expert pool across layers, with layer-specific backbone weights. Chunk-start reads and one aggregated update per chunk preserve causality. The matched TTT-MoE control isolates sharing from routing.

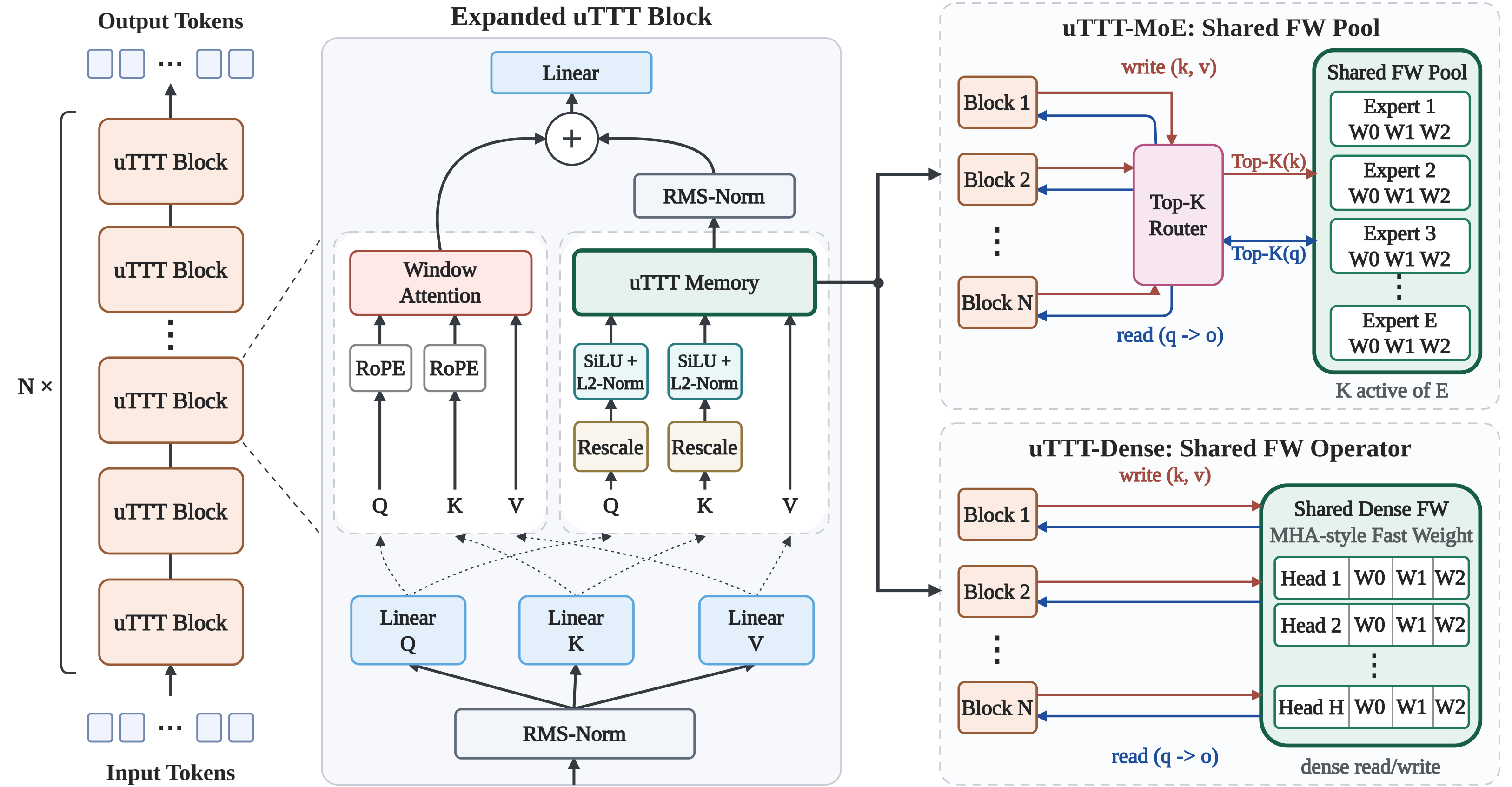
### 2 Sharing improves quality and efficiency

At equal state and active compute, uTTT-MoE gains 2.6 / 2.1 RULER points over TTT-MoE at 124M / 760M. In view synthesis, MoE / Dense sharing adds 0.92 / 0.76 dB at view 23. Dense sharing keeps per-layer compute with 1/8 the state. Aggregating writes also enables up to 1.39× faster training than TTT-MoE.

### 3 A unified memory recurrence across time and depth

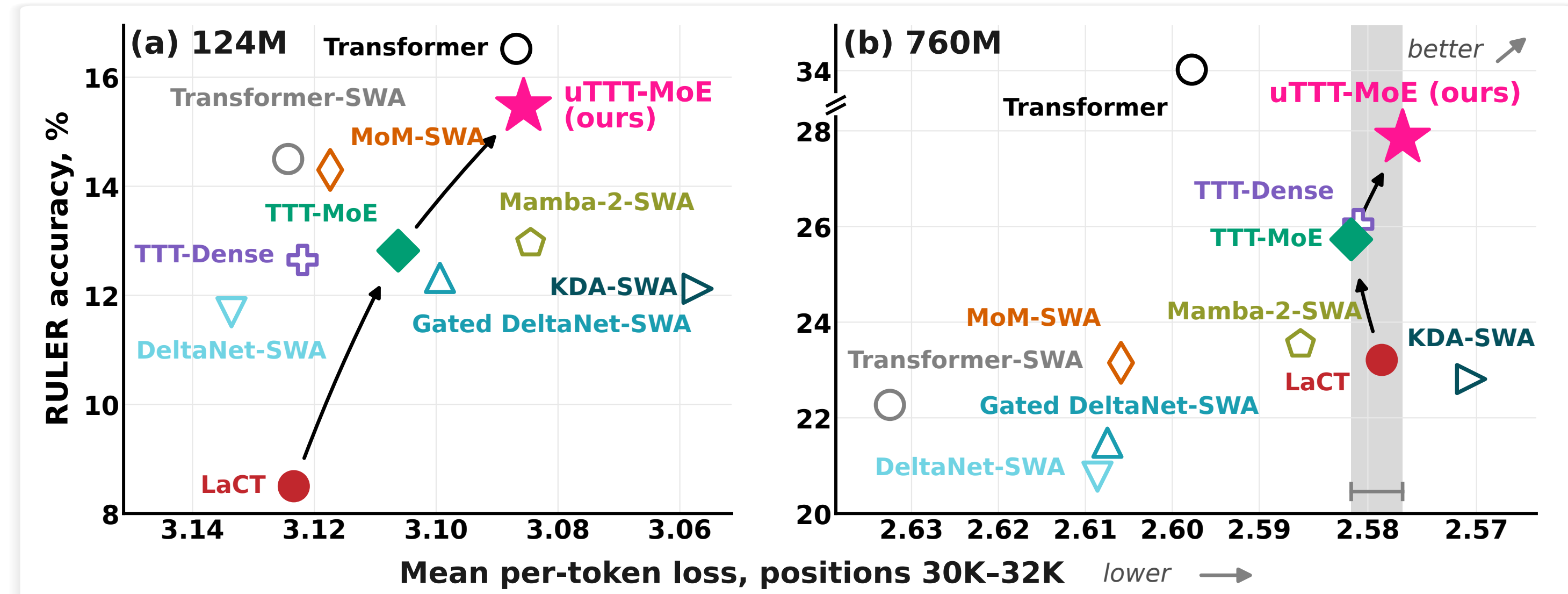
Shared memory evolves across both time and depth, with layer-private TTT as a partitioned special case. A shallow read can use earlier deep-layer writes, and the task loss can backpropagate through those writes. In trained MoE models, probes detect these routes. Removing cross-layer reads at inference or cutting cross-depth gradients during training degrades performance.

## uTTT: shared fast weights across layers



## Long-context language modeling

124M / 760M models; 32K training context; 4K chunks and local attention. PTL (nat): ≥30K on Books3. RULER (%): mean over 4K, 8K, 16K and 32K.



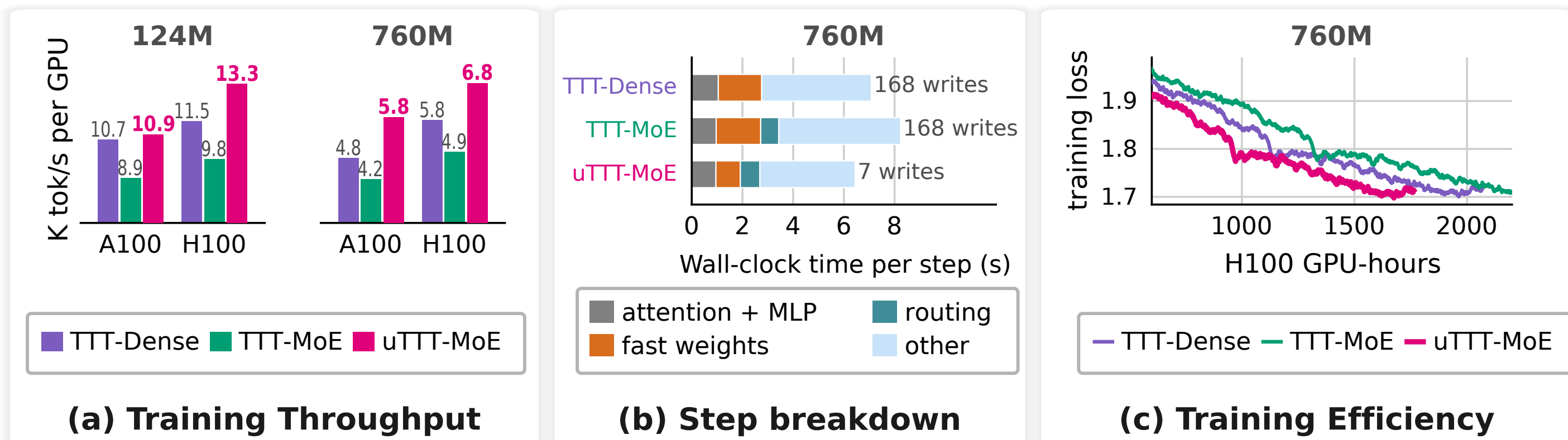
| Model              | 124M          |              | 760M          |              |
|--------------------|---------------|--------------|---------------|--------------|
|                    | PTL ↓         | RULER ↑      | PTL ↓         | RULER ↑      |
| Transformer†       | 3.0869        | 16.52        | 2.5978        | 34.16        |
| Transformer-SWA    | 3.1243        | 14.50        | 2.6325        | 22.27        |
| Gated DeltaNet-SWA | 3.0994        | 12.32        | 2.6075        | 21.48        |
| DeltaNet-SWA       | 3.1336        | 11.69        | 2.6086        | 20.77        |
| KDA-SWA            | <b>3.0571</b> | 12.12        | <b>2.5705</b> | 22.81        |
| Mamba-2-SWA        | 3.0845        | 12.96        | 2.5862        | 23.54        |
| MoM-SWA            | 3.1174        | 14.30        | 2.6059        | 23.15        |
| LaCT (published)   | 3.1234        | 8.50         | 2.5787        | 23.21        |
| TTT-Dense          | 3.1219        | 12.65        | 2.5809        | 26.05        |
| TTT-MoE            | 3.1062        | 12.82        | 2.5815        | 25.73        |
| <b>uTTT-MoE</b>    | <b>3.0857</b> | <b>15.46</b> | <b>2.5768</b> | <b>27.85</b> |

**+2.64 / +2.12 RULER points vs. TTT-MoE at equal state and compute**

Bold: best bounded-state result. † Full attention retrieves better with a growing KV cache.

## Measured training efficiency

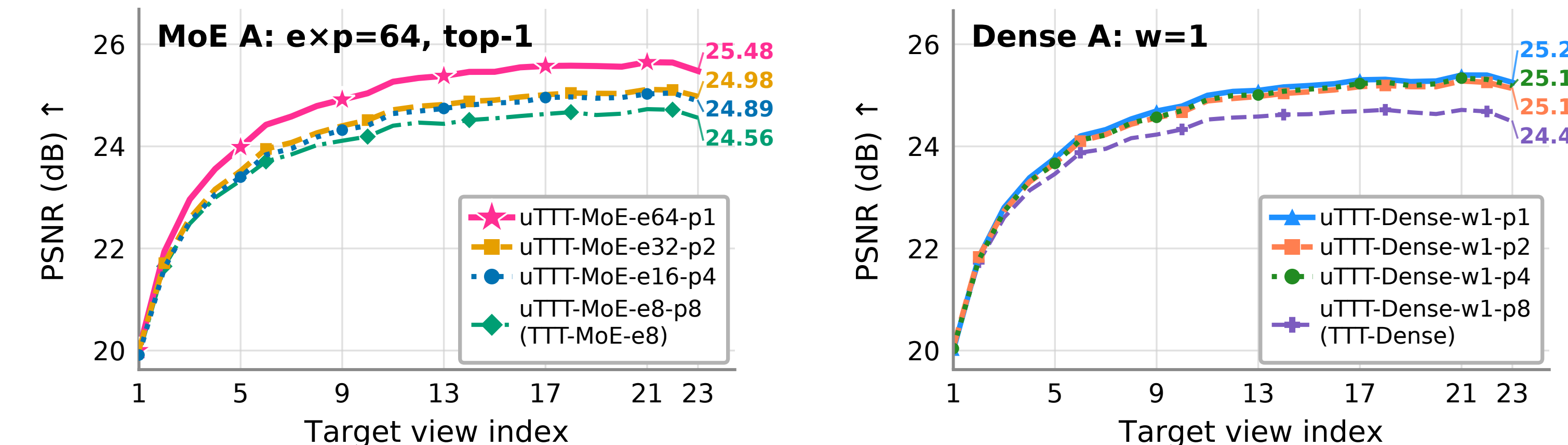
32K context, batch 1/GPU. (a) 8 A100/H100 GPUs; (b) 1 A100; (c) H100 GPU-hours.



**Up to 1.39× throughput; 27% fewer GPU-hours to matched loss vs. TTT-MoE.**

## Novel view synthesis: MoE and Dense

Objaverse → GSO; within-view attention; 11 train / 23 eval writes. Mean metrics: views 1–23 per object, then 1,019 objects. PSNR in dB.



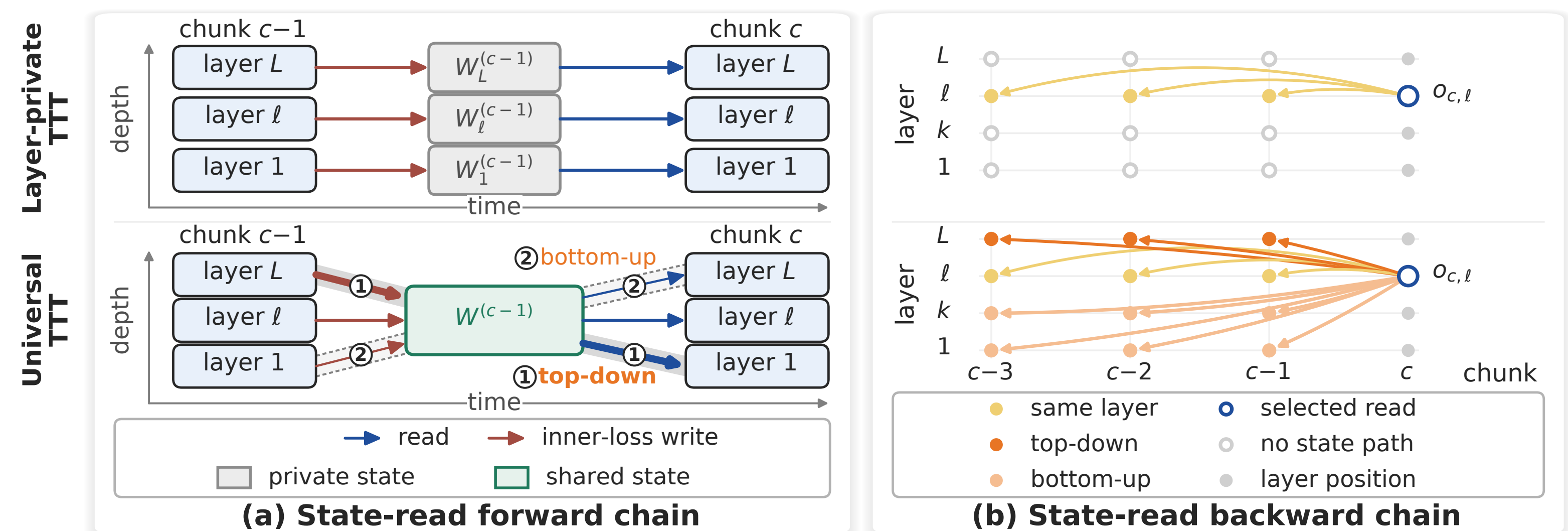
| Model                     | Mean PSNR ↑  | View-23 PSNR ↑ | SSIM ↑       | LPIS ↓       |
|---------------------------|--------------|----------------|--------------|--------------|
| Gated DeltaNet            | 24.32        | 25.01          | 0.856        | 0.178        |
| DeltaNet                  | 23.94        | 24.55          | 0.851        | 0.185        |
| TTT-MoE-e8 (p=8)          | 23.90        | 24.56          | 0.850        | 0.186        |
| <b>uTTT-MoE-e64 (p=1)</b> | <b>24.68</b> | <b>25.48</b>   | <b>0.860</b> | <b>0.173</b> |

**8 → 1 pools: +0.92 dB MoE / +0.76 dB Dense at view 23**

Sharing sweeps fix FW compute/layer. MoE: same state (64 experts, top-1).

Dense: 1/8 state at unchanged read/write width. p = number of pools.

## Cross-depth memory: forward and backward routes



Backward arrows trace state-chain gradients through past updates, with queries and coefficient producers fixed. Hollow nodes mark absent state-chain paths.

MoE evidence: read-output gradients reach 77/77 earlier deep writes (TTT-MoE: 0/77). Cutting the layer-8 → layer-3 route costs 0.69 dB on GSO.